

Analyzing the Impact of Pit Stops and Circuit Characteristics on Formula 1 Race Outcomes

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Figure 1: image from Formula 1 Official Website

ABSTRACT

Formula 1 (F1) racing is a high-stakes sport where race outcomes are influenced by multiple factors, including driver performance, pit stop efficiency, and circuit characteristics. This study aims to analyze the impact of pit stop strategies and circuit attributes on race results. Using data from F1 races between 2020 and 2024, we apply machine learning models to determine how pit stop timing, number of stops, and circuit-specific elements influence race outcomes. Our results suggest that optimizing pit stop strategies can significantly enhance a driver's final position, with certain circuits requiring more aggressive strategies than others.

1. INTRODUCTION

Formula 1 is a sport that combines advanced engineering with real-time strategy. Teams must make split-second decisions on pit stops and tire changes, balancing speed and endurance. The goal of this study is to determine the impact of pit stop strategies and circuit attributes on race results. By understanding these relationships, teams can optimize race strategies, improve performance, and increase their chances of success.

2. RELATED WORK

Previous studies in motorsports analytics have explored predictive modeling for race outcomes,

analyzing driver skill, vehicle aerodynamics, and team strategies. However, limited work has been done on quantifying the impact of pit stop efficiency and circuit characteristics on final race positions. Our research builds on this by incorporating machine learning techniques to analyze these factors.

3. DATASET OVERVIEW

We use multiple datasets covering F1 races, drivers, and pit stops. Key datasets include:

- F1 Races (2000-2024): Details of each Grand Prix, including circuit ID, date, and results.
- Drivers Dataset: Information on drivers, including names and teams.
- Pit Stops Dataset: Records of pit stop times, number of stops, and their effects on performance.

4. METHODOLOGY

Our approach involves data preprocessing, feature engineering, and predictive modeling.

Feature Engineering: We create features such as average pit stop duration, number of pit stops per driver, and circuit difficulty levels.

Machine Learning Models: We employ XGBoost models to predict race outcomes.

Dimensionality Reduction: Using PCA and t-SNE, we visualize feature relationships and importance.

5. MAPPING CIRCUIT IDs TO GRAND PRIX NAMES

5.1 Background & Motivation

A major challenge in working with raw Formula 1 race data was that circuits were identified only by numerical IDs, lacking corresponding Grand Prix names. This made it difficult to track circuit appearances by season, compare driver performance across races, and incorporate

track-specific factors into predictive models. Manually mapping each circuit ID was inefficient, so we leveraged ChatGPT to research and verify Grand Prix names, streamlining the process while improving accuracy.

5.2 Step 1- Extracting Race Dates & Circuit IDs for Each Season

Before mapping circuits, we first needed to understand which circuits were used in each year. This involved:

1. Filtering race data by year (2000–2024) to retrieve race instances.
2. Grouping by race date and circuit ID to capture unique race events for each season.
3. Compiling the extracted data into a structured reference table, ensuring that circuit appearances were correctly recorded.

This step was essential for maintaining seasonal accuracy, as the Formula 1 calendar changes over time, with circuits being added, removed, or rotated between seasons.

5.3 Step 2 - Mapping Circuit IDs to Grand Prix Names

After structuring a dataset of circuit IDs and race dates, we mapped each ID to its corresponding Grand Prix name. To avoid the inefficiency of manual research, we saved the cleaned data to a CSV file and used ChatGPT to gather and verify the mappings. This approach streamlined the process, improved accuracy through cross-referencing, and ensured proper labeling of all circuits, including temporary or discontinued ones. The final mapping allowed for more intuitive circuit analysis.

5.4 Step 3 - Integrating Circuit Names into the Dataset

After mapping Grand Prix names to circuit IDs, we integrated them into our dataset, ensuring each track was recognizable. This transformation improved historical circuit tracking, enhanced race and driver performance analysis, and provided a structured reference for machine learning applications.

5.5 Conclusion

This mapping process transformed raw circuit IDs into meaningful track names, enabling:

1. More intuitive data analysis, with recognizable track names instead of arbitrary IDs.
2. Better driver-track performance modeling, as we could now associate historical performances with specific circuits.
3. Seamless integration into predictive models, allowing us to incorporate track characteristics into podium prediction analysis.

By automating this research step with ChatGPT, we optimized the workflow and ensured a high level of accuracy in circuit naming, setting the foundation for further race analytics.

6. DRIVER-TRACK RELATIONSHIP MODELING USING WORD EMBEDDINGS AND CLUSTERING

6.1 Background and Motivation

Understanding driver-circuit relationships is essential in Formula 1 analysis, but traditional methods often overlook patterns like track specialization and adaptability. To address this, we used Word2Vec embeddings to model driver-track relationships, followed by PCA and K-Means clustering for visualization. This

approach quantified circuit similarities based on driver behavior and identified potential track specializations. By applying Word2Vec, a technique from natural language processing, we captured latent connections between tracks based on driver frequency, enhancing race analysis

6.2 Preprocessing and Data Preparation

Before training the Word2Vec model, we preprocessed the dataset for compatibility with the embedding framework. Circuit and driver IDs were converted to strings, and each driver's race history was aggregated into a sequential list, treating circuits as "words" in a "sentence." This allowed Word2Vec to learn transition patterns across tracks.

We trained a 50-dimensional Word2Vec model using the Skip-gram approach (sg=1) to capture rare circuit-driver pairings. With a window size of 5 and 50 training epochs, the model balanced local and global track associations. Testing involved retrieving similar circuits based on embeddings, providing an early assessment of its effectiveness in capturing meaningful track relationships.

6.3 Dimensionality Reduction and Visualization

To visualize the high-dimensional (50D) Word2Vec embeddings, we used Principal Component Analysis (PCA) to project them onto a two-dimensional space while preserving variance. The PCA visualization revealed clusters where circuits with similar racing characteristics grouped together, and drivers positioned near specific tracks suggested potential track specialists. This provided an initial insight into driver-circuit relationships, setting the stage for deeper analysis.

6.4 Clustering Circuits and Drivers Using K-Means

To formalize the relationships observed in the PCA plot, we applied K-Means clustering to the reduced embeddings, grouping the dataset into five clusters. This revealed distinct circuit-driver relationships by:

- **Identifying Circuit Similarities** – Tracks with shared characteristics, such as high-speed circuits (Monza, Silverstone) or technical street circuits (Monaco, Singapore, Baku), were grouped together.
- **Classifying Drivers** – Some drivers clustered around specific track types, indicating specialization, while others were more evenly distributed, suggesting adaptability.

The results were visualized in a color-coded scatter plot, providing an interpretable view of track and driver relationships. This clustering laid the foundation for feature engineering in predictive modeling.

6.5 Findings and Interpretation

The application of Word2Vec embeddings, PCA visualization, and K-Means clustering revealed key insights into driver-track relationships:

- **Track Characteristics** – High-speed circuits formed distinct clusters separate from street circuits, confirming that the model captured meaningful track similarities.
- **Driver Specialization** – Some drivers showed clear track preferences, while others demonstrated adaptability, providing quantifiable trends for predictive modeling.
- **Enhanced Race Predictions** – By assigning drivers and circuits to clusters,

track specialization became a valuable feature in machine learning models.

These insights were later integrated into the podium prediction framework, strengthening the understanding of historical driver performance on similar circuits.

7. FEATURE ENGINEERING FOR PODIUM PREDICTION

7.1 Background and Motivation

To build an effective podium prediction model, we engineered features that captured key race performance factors beyond simple race results. Since podium finishes depend on variables like driver-track performance, weather conditions, and race dynamics, we developed:

- **Driver-Track Performance Features** – Quantifying a driver's historical success at different circuits.
- **Weather Impact Features** – Assessing a driver's likelihood of finishing a race under various weather conditions.

These features enabled the model to learn how past performances influence future race outcomes, enhancing its predictive accuracy.

7.2 Driver-Track Performance Features

Since different drivers perform differently on various circuits, we computed historical driver statistics per circuit to quantify track-specific performance.

Podium Rate Calculation

A new feature, `podium_rate`, was created by computing the percentage of races where a driver finished in the Top 3 at a specific circuit. This metric provides insight into track specialization and highlights drivers who consistently perform well at certain venues.

Average Points Per Circuit

The mean points scored per circuit (`avg_points`) was calculated for each driver.

This metric accounts for consistent high finishes, even if they did not always result in podiums.

7.3 Weather Impact Features

Weather significantly impacts Formula 1 race outcomes, with rain increasing DNFs and altering strategies. To quantify this, we engineered features such as **DNF rates in dry vs. wet conditions** to model a driver's likelihood of finishing a race and **average points in different weather conditions** to assess adaptability. Since weather is a binary variable (dry = 0, wet = 1), we restructured the dataset to create separate columns for dry and wet race statistics, integrating them into the main dataset. These features allowed the model to account for external conditions when predicting podium potential.

7.4 Data Cleaning and Final Preparation for Modeling

After feature engineering, we cleaned the data to ensure consistency by renaming duplicate columns, removing unnecessary duplicates, and verifying feature integrity. This ensured that all race entries contained structured, meaningful predictive features for the podium prediction model.

8. MODEL TRAINING AND FEATURE WEIGHT ADJUSTMENTS FOR PODIUM PREDICTION

After engineering features on driver-track relationships, weather impact, and race outcomes, we trained an XGBoost model for podium prediction. However, standard race features alone were insufficient, as grid position and historical podium rates dominated

predictions. To improve balance, we adjusted feature weights and incorporated clustering data to evaluate the impact of track specialization, enhancing the model's accuracy.

8.1 Incorporating Clustering for Track Specialization

To incorporate track specialization into the model, we integrated clustering data from the Word2Vec-based embeddings. We ensured each driver had a cluster assignment, calculated average podium rates per cluster, and merged these rates into the dataset. This allowed the model to assess whether drivers specializing in certain track types had a higher probability of podium finishes.

8.2 Feature Weight Adjustments for Model Training

During initial model training, grid position overly influenced predictions, favoring front starters. To balance this, we applied an inverse transformation to normalize grid position, increased the weight of historical podium success and average points per race by 1.5x, and emphasized wet-weather DNF rates by 1.3x to account for unpredictability. These adjustments ensured a more accurate reflection of driver skill and adaptability.

8.3 Model Training with Adjusted Features

After applying feature weight adjustments, we trained the XGBoost model by first handling missing data, filling gaps with median values to avoid bias. The dataset was then split 80/20 for training and testing to ensure robust evaluation. Hyperparameter optimization involved tuning **learning_rate**, **max_depth**, and **n_estimators**, with the best results achieved using **learning_rate = 0.05**, **max_depth = 7**, **n_estimators = 500**, **subsample = 0.8**, and

colsample_bytree = 0.8 to balance accuracy and recall while preventing overfitting.

8.4 Challenges Encountered During Model Training

Early model iterations faced three key challenges. **Over-reliance on grid position** led to podium predictions favoring the top three starters, which was mitigated by normalizing grid position and boosting other performance-related features. **Class imbalance** made podium predictions difficult, as only three out of twenty positions finish on the podium; while SMOTE was tested, carefully adjusting feature weights proved more effective. Lastly, **track specialization had limited impact**, with general driver skill playing a larger role, so while track clusters were retained, their influence was lower than expected.

8.5 Model Performance and Evaluation

After implementing feature weight adjustments and optimizing hyperparameters, the final XGBoost model achieved an accuracy of 89.23%.

Metric	Score
Accuracy	89.23%
Precision	0.61
Recall	0.56
F1-Score	0.59

The model achieved **89.23% accuracy**, effectively distinguishing podium from non-podium finishes. **Precision (0.61)** improved, reducing false positives, but **recall (0.56)** remained low, indicating some missed podiums. The F1-score (0.59) suggests room for improvement.

8.6 Key Observations

- Reduced Grid Bias – Feature weighting adjustments balanced grid position influence.
- Stronger Driver Performance Impact – Historical podium rates and points improved predictions.
- Limited Track Specialization Effect – Clusters were useful but not as predictive as past results.
- Moderate Influence of Weather & Pit Stops – These factors added unpredictability but were secondary to performance history.

Overall, podium finishes are driven by consistent performance trends, while strategy elements like pit stops and weather add variability.

9. MODEL VALIDATION USING HISTORICAL RACE DATA

To validate the podium prediction model, we tested it against real-world results, beyond standard accuracy metrics. The **2024 Las Vegas Grand Prix** was chosen as a test case because it had normal conditions, used recent season data, and provided an unbiased evaluation. The goal was to assess whether the drivers with the highest predicted podium probabilities actually finished in the top three.

9.1 Model Predictions vs. Actual Race Results

The model’s predictions were compared to the actual finishing positions in the Las Vegas GP.

Driver	Predicted Podium Probability	Actual Finish Position
RUS	0.862	1st
HAM	0.833	2nd
SAI	0.465	3rd

LEC	0.166	4th
VER	0.0006	5th
NOR	0.023	6th
PIA	0.013	7th
TSU	0.0001	9th
GAS	0.0005	DNF
ALB	0.0001	DNF

3. Evaluation of Model Performance

The model performed well, accurately predicting the top three finishers—Russell, Hamilton, and Sainz—as having the highest podium probabilities. It correctly ranked lower-probability drivers like Tsunoda, Gasly, and Albon, who finished outside the top positions or retired. However, Verstappen’s unusually low probability (0.0006) despite finishing fifth suggests a potential bias in the model.

The results demonstrate strong reliability in identifying top-performing drivers, confirming that grid position normalization worked effectively. Hamilton, who started from P10, still received a high podium probability due to his strong historical performance. However, Verstappen’s low probability highlights the need for further refinements to ensure the model accurately captures top-tier driver consistency across different circuits.

10. INITIAL PREDICTIONS FOR 2025 AUSTRALIAN GRAND PRIX

After validating the model on past data, we tested it on the **2025 Australian Grand Prix** to assess real-world performance. This race was chosen for its relevance, normal conditions, and as a baseline before integrating weather and pit stop data. However, since the model assumed dry conditions, it did not account for the race’s

wet weather impact. The effect of incorporating weather data will be analyzed in the following sections.

10.1 Feature Selection for Prediction

The model predicted podium probabilities using key features, including **normalized grid position**, **driver performance metrics** (average points, podium rate, and DNF rates in dry/wet conditions), and **weighted adjustments** to emphasize podium history (1.5x) and wet race DNF rates (1.3x, though inactive for this dry-race assumption). These processed features were then input into the trained XGBoost model.

10.2 Model Predictions for the 2025 Australian Grand Prix

After processing the race data and applying the trained model, the podium probabilities for each driver were generated and sorted in descending order.

Driver	Predicted Podium Probability
HAM (P8)	0.7217 (72.2%)
NOR (P1)	0.6752 (67.5%)
PIA (P2)	0.5651 (56.5%)
RUS (P4)	0.1677 (16.8%)
SAI (P10)	0.0085 (0.8%)
LEC (P7)	0.0082 (0.8%)
VER (P3)	0.0007 (0.07%)
TSU (P5)	0.0004 (0.04%)
GAS (P9)	0.0002 (0.02%)
ALB (P6)	0.0001 (0.01%)

However, it is important to highlight that these predictions were made under the assumption of dry conditions, while the actual 2025 Australian Grand Prix was a wet race. This discrepancy will be explored in later sections when we integrate

weather conditions and pit stop data into the model.

10.3 Incorporating Weather and Pit Stops

At this stage, weather conditions and pit stops were not yet integrated. The next step involves refining predictions by adding actual race conditions, incorporating pit stop strategies, and comparing results for accuracy improvements.

To enhance the model, we introduced **weather conditions (wet vs. dry)** and **total pit stops per driver**, as these factors significantly impact race outcomes. Wet weather increases unpredictability and DNF rates, while pit stop timing affects track position. These additions aim to improve prediction accuracy and better reflect real-world race dynamics.

We integrated **pit stop data** by merging total stops per driver into the dataset, filling missing values with zero for non-stoppers. **Weather conditions** were added as a binary feature (1 = wet, 0 = dry) to differentiate race conditions.

After updating the **XGBoost model**, accuracy remained stable at **89.68%**, with a slight increase in **podium precision (0.64)**, though recall (**0.56**) remained unchanged. **Pit stops had a moderate impact** but did not outweigh key factors like grid position and average points. **Weather effects varied by driver**, with some showing consistently lower DNF rates in wet conditions.

Pit stops contributed to accuracy but were not dominant, and the model still underestimates weather's impact. Future refinements may focus on **pit stop timing** rather than total stops and further tuning weather influence on predictions.

Integrating weather conditions and pit stops enhanced the model by incorporating race strategy and adaptability, making predictions

more reflective of real-world conditions. While accuracy remained stable, the model now considers external factors beyond driver history and grid position. Next, we apply this updated model to the **2025 Australian Grand Prix** and compare results with earlier predictions that excluded weather and pit stop data.

11. FEATURE IMPORTANCE ANALYSIS

Understanding feature importance ensures the model correctly weights key race factors. By using XGBoost's built-in ranking, we assessed the impact of **driver performance, grid position, weather, and pit stops** on podium predictions.

- **Driver performance (avg_points, podium_rate) and grid position remain the strongest predictors**, confirming that historical success heavily influences podium finishes.
- **Pit stops ranked 4th in importance**, contributing to accuracy but not outweighing core performance metrics.
- **Weather conditions had a moderate impact**, affecting some predictions but not overriding driver skill.
- **Grid position bias was reduced**, ensuring that other factors played a meaningful role.

The model now effectively balances **historical results, strategy, and external conditions**, improving its predictive capability.

12. FINAL MODEL PREDICTIONS FOR 2025 AUSTRALIAN GRAND PRIX

With the updated model incorporating **weather conditions and pit stops**, we predicted podium probabilities for the **2025 Australian Grand Prix**, a wet-weather race with unpredictable conditions.

12.1 Model Performance vs. Actual Results:

- **Correct Predictions:** The model accurately ranked Norris as the top podium contender, and he won the race. Russell, though assigned a lower probability, finished 3rd.
- **Misses & Insights:** Verstappen, predicted with an extremely low probability (0.0006), finished 2nd, suggesting the model underestimates his adaptability to wet conditions. Hamilton (high probability) finished 10th, possibly due to adjusting to his new Ferrari.
- **Race Incidents:** Piastri, predicted as a podium finisher, was running P2 before a late spin dropped him to 9th, an unpredictable event the model cannot account for.

Driver	Predicted Podium Probability	Actual Finish Position
NOR	0.7842	1st ✓
HAM	0.7167	10th ✗
PIA	0.6426	9th ✗
RUS	0.0774	3rd ✓
SAI	0.0049	DNF
LEC	0.0174	8th
VER	0.0006	2nd ✗
TSU	0.0003	12th
GAS	0.0002	11th
ALB	0.0001	5th ✗

12.2 Justification of Model Performance and Observations

- **Weather & pit stops improved accuracy**, but **driver adaptability to wet conditions needs better modeling**.
- **Real-time adjustments** may enhance predictions as the **2025 season progresses**, refining the weight of **pit stops, weather, and track conditions**.
- **Continuous updates** will help the model adapt to team and driver performance trends, making future predictions more accurate.

The model serves as a **strong baseline** for future races, with ongoing refinements improving its predictive power as more data becomes available.

13. Conclusion & Future Work

This study highlights the importance of pit stops and circuit characteristics in F1 race outcomes. Our findings suggest that teams should optimize pit stop strategies based on circuit-specific attributes. Future work will explore real-time predictive analytics, integrating weather conditions and in-race telemetry data to enhance predictive accuracy.

Github

Peggy Tang:

<https://github.com/rsm-p6tang/Impact-of-Pit-Stops-and-Circuit-Characteristics-on-Formula-1-Race-Outcomes>

Carolyn Su:

<https://github.com/rsm-cas024/MGTA-415-final-project>